# CNC STEPCRAFT D600

## *CNC Use*

**Please do not use the CNC without training by Raph or Vu before. It's not super difficult but there are quite a few mistakes that can be made.**

Safety:

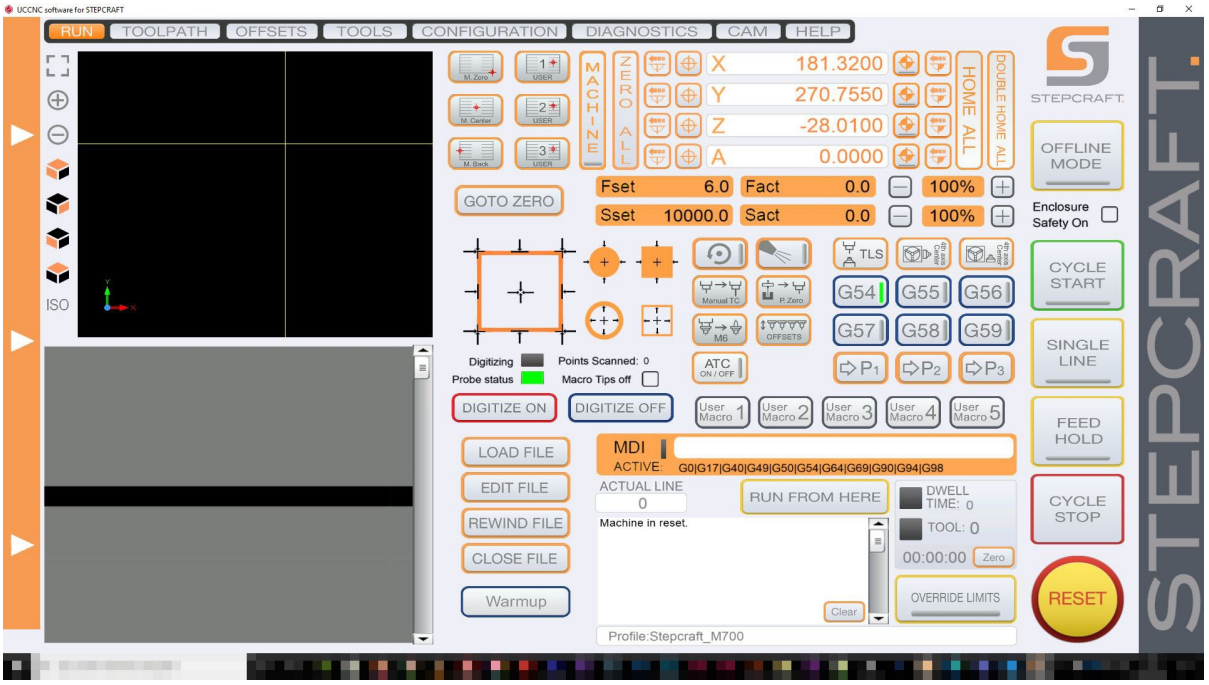
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| Closed enclosure during carbon cutting |
| Be in the room during cutting |
| Properly vacuum after cutting using the filtering system. |
| Wear latex gloves (to avoid irritation) to remove the parts/clean. |

## *Cutpaths generation*

Please find the full setup guide here: **PDF Guide**

## *Cutting*

1. Start the UCCNC software
2. Home all axis
3. Stick your carbon plate to the top of a sacrificial, flat, wooden/plastic layer. Very important: the **entirety** of the zone that will be cut needs to be covered in double-sided tape. Make sure there are no folds/impurities to keep everything flat
4. Fix this assembly in the machine with the 2 transversal metal fixtures. Firmly screw down.
5. Zero X and Y to the start point of your cut.
6. Zero Z to a point **5 cm above the plate**. Then **load file** and **cycle start** (launch the cutting job) **in the air** to verify that the tool follows the wanted profile.
7. Close the enclosure completely and **turn on the spindle** or else you will damage the tool.
8. Zero Z to either the top or the bottom of the carbon plate (depending on the fusion setup). **Do this slowly**. When you start seeing the first sign of carbon dust it means you are touching the plate/support and that is your 0.
9. Lower the slider right of **Fact** from 100% to 5% (see below).
10. Start cutting
11. During cutting, gradually increase the speed (next to Fact) to 10% for the 1 mm endmill, 20% for the 2 mm endmill, 30% for the 3 mm endmill (given 0.8 mm passes in carbon).




## *After cutting*

1. Check that the cut went throught the plate. If not quite, move the machine to 0, lower the Z zero slightly by 0.5 mm and cut again.
2. Vacuum clean the part before removing. Wear gloves
3. Unscrew the support rods
4. Unstick the carbon plate and cuts from the sacrificial layer. Spraying some acetone on helps.
5. **Remove any tape** from the left over carbon plate.
6. Unplug machine and turn off computer.

## *Files*

https://wiki.epfl.ch/lis/documents/equipment/cnc-stepcraft/stepcraft uccnc.cps

https://wiki.epfl.ch/lis/documents/equipment/cnc-stepcraft/CarbonEndMills.tools

-- instructions by Raph Zufferey